
Open Issues in `std::execution::task`

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R1: March 2026 (prior to Croydon meeting)

R0: February 2026 (pre-Croydon mailing)

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Abstract

`std::execution::task` ([P3552R3^{\[1\]}](#)) has open issues identified by national ballot comments, LEWG issues, and published papers. This paper classifies each issue by whether it can be resolved after C++26 ships or whether shipping forecloses the fix.

Revision History

R1: March 2026 (prior to Croydon meeting)

- Complete rewrite as an informational classification of open issues.

R0: February 2026 (pre-Croydon mailing)

- Original analysis of structural gaps. See [P4007R0](#)^[2].
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1. Disclosure

The authors developed [P4007R0](#)^[2] (“Senders and Coroutines”). The classification below holds regardless of whether any alternative design exists.

2. Fixed After Ship

`task`’s `promise_type` is a class template instantiated in user code. Its `operator new`, allocator selection, stop token storage, environment forwarding, and destruction ordering can change between standard revisions without binary incompatibility. These issues are fixable post-ship: [Unusual Allocator Customisation](#)^[3], [Flexible Allocator Position](#)^[3], [Shadowing The Environment Allocator](#)^[3], [Stop Source Always Created](#)^[3], [Stop Token Default Constructible](#)^[3], [Task Not Actually Lazily Started](#)^[3], [Frame Destroyed Too Late](#)^[3], [No Default Arguments](#)^[3], `unhandled_stopped` `Not noexcept`^[3], [Environment Design Inefficient](#)^[3], [Non-Sender Awaitables Unsupported](#)^[3], [Future Language Feature Could Avoid `co\_yield`](#)^[3], [No TLS Capture/Restore Hook](#)^[3], `return_value / return_void` [Have No Specification](#)^[3], `co_return { args... }` [Unsupported](#)^[3], `change_coroutine_scheduler` [Requires Assignable Scheduler](#)^[3], [Sender-Unaware Coroutines Cannot `co\_await a task`](#)^[3], [Missing Rvalue Qualification](#)^[3], [Parameter Lifetime Is Surprising](#)^[4], [No Protection Against Dangling References](#)^[4], `co_yield with_error` [Is Clunky](#)^[4], `co_await schedule(sch)` [Is an Expensive No-Op](#)^[4], [Coroutine Cancellation Is Ad-Hoc](#)^[4].

`affine_on` semantics, rescheduling behavior, and algorithm customisation are specification-level concerns. Tightening requirements or adding default implementations does not change any published interface. These issues are fixable post-ship: `affine_on` [Default Implementation Lacks Specification](#)^[3], `affine_on` [Semantics Not Clear](#)^[3], `affine_on` [Shape May Not Be Correct](#)^[3], `affine_on` [Shouldn’t Forward Stop Requests](#)^[3],

`affine_on` Customisation For Other Senders^[3], Starting a `task` Reschedules Unconditionally^[3], Resuming After a `task` Reschedules Unnecessarily^[3], `bulk` vs. `task_scheduler`^[3], No Completion Scheduler^[3], `with_awaitable_senders` Unused^[3].

3. Not Fixable Post-Ship

The issues in this section are items where shipping forecloses the fix.

Issue	References	Fixed
Allocator Timing	P3980R0 ^[5] , P3796R1 ^[3] , LWG 4356 ^[6] , US 254-385 ^[7]	no
Allocator Propagation	P3980R0 ^[5] , P3796R1 ^[3]	no
Error Return	P3950R0 ^[8] , P3801R0 ^[4] , P1713R0 ^[9]	no
Symmetric Transfer	P2583R3 ^[10] , US 246-373 ^[11] , LWG 4348 ^[12] , P3801R0 ^[4] , P3796R1 ^[3]	no

- **Allocator Timing.** `task` does not allow the user to specify an allocator at the launch site for coroutine frame allocation. The coroutine frame is allocated by `operator new` at the call site, before any sender `connect` / `start` machinery runs. The receiver's environment - which carries the allocator - is not yet available at that point. The `allocator_arg` mechanism in P3552R3^[1] works around this by requiring the allocator in the coroutine's parameter list, but that locks in a caller-specified approach and forecloses environment-based injection.
- **Allocator Propagation.** `task` does not propagate the frame allocator to child tasks. Each child task allocates its frame independently. Shipping without propagation locks in a design where every launch site must specify its allocator explicitly, foreclosing transparent propagation through the coroutine call tree.
- **Error Return.** `task` requires `co_yield with_error(e)` to propagate an error to the caller. `co_return` cannot carry an error value because `return_value` and `return_void` are mutually exclusive in the current coroutine specification. Shipping this interface locks in the `co_yield` mechanism and forecloses `co_return`-based error propagation, which would require a language change.
- **Symmetric Transfer.** The completion functions (`set_value` , `set_error` , `set_stopped`) and `start()` return `void` , providing no channel to propagate a `coroutine_handle<>` . When a sender completes synchronously, the receiver calls `handle.resume()` on the caller's stack, adding a frame per completion

with no upper bound. Shipping this protocol forecloses the `coroutine_handle<>`-returning completion protocol that would enable symmetric transfer.

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References

Papers, issues, and ballot comments referenced in this document.

WG21 Papers

1. **P3552R3** - “Add a Coroutine Task Type” (Dietmar Kühl, Maikel Nadolski, 2025). <https://wg21.link/p3552r3>
2. **P4007R0** - “Senders and Coroutines” (Vinnie Falco, Mungo Gill, 2026). <https://wg21.link/p4007r0>
3. **P3796R1** - “Coroutine Task Issues” (Dietmar Kühl, 2025). <https://wg21.link/p3796r1>
4. **P3801R0** - “Concerns about the design of `std::execution::task`” (Jonathan Müller, 2025).
<https://wg21.link/p3801r0>
5. **P3980R0** - “Task’s Allocator Use” (Dietmar Kühl, 2026). <https://isocpp.org/files/papers/P3980R0.html>
6. **LWG 4356** - “`connect()` should use `get_allocator(get_env(rcvr))`”. <https://cplusplus.github.io/LWG/issue4356>
7. **US 254-385** - C++26 NB ballot comment. <https://github.com/cplusplus/nbballot/issues/960>
8. **P3950R0** - “`return_value` & `return_void` Are Not Mutually Exclusive” (Robert Leahy, 2025).
<https://wg21.link/p3950r0>
9. **P1713R0** - “Allowing both `co_return;` and `co_return value;` in the same coroutine” (Lewis Baker, 2019).
<https://wg21.link/p1713r0>
10. **P2583R3** - “Symmetric Transfer and Sender Composition” (Mungo Gill, Vinnie Falco, 2026).
<https://isocpp.org/files/papers/P2583R3.pdf>
11. **US 246-373** - C++26 NB ballot comment. <https://github.com/cplusplus/nbballot/issues/948>
12. **LWG 4348** - “`task` doesn’t support symmetric transfer”. <https://cplusplus.github.io/LWG/issue4348>
13. **P2300R10** - “`std::execution`” (Michał Dominiak et al., 2024). <https://wg21.link/p2300r10>

Other

14. **C++ Working Draft** - (Richard Smith, ed.). <https://eel.is/c++draft/>